

Dataset Complexity via Multi-Scale Score Distance

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April 4, 2026

Abstract

We propose $\mathcal{C}(\mathcal{X}) = \log \det(I + K)$ as a scalar index of dataset complexity, where $K_{ij} = \exp(-\lambda D(x_i, x_j)^2)$ and D is a multi-scale distance derived from the probability flow ODE of a pretrained diffusion model.

Given a score function $s(x, t) = \nabla_x \log p_t(x)$ learned by a diffusion model, each data point x defines a deterministic trajectory $z_x(t)$ through the probability flow ODE. This trajectory maps x from data space toward the noise prior, passing through intermediate representations that encode the semantic neighbourhood structure of x at every scale. Recording the trajectory at L time points produces a feature map $\Phi(x) = (\sqrt{w_1} z_x(t_1), \dots, \sqrt{w_L} z_x(t_L)) \in \mathbb{R}^{DL}$; the distance $D(x_i, x_j) = \|\Phi(x_i) - \Phi(x_j)\|_{\mathbb{R}^{DL}}$ measures how differently two data points traverse the diffusion process across all scales.

Because D is the Euclidean distance in the flat space $\mathbb{R}^{DL}$, it is conditionally negative definite, and $K = \exp(-\lambda D^2)$ is positive semi-definite by Schoenberg’s theorem. This gives $\mathcal{C}$ the full suite of desirable properties: permutation invariance, bounded monotonicity, sub-additivity, submodularity, and Lipschitz stability. All proofs are self-contained.

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1 Introduction

As the scale of datasets used to train generative models reaches trillions of tokens and billions of images, the question of whether dataset size is a sufficient proxy for “information content” becomes critical. Current paradigms frequently assume that overall complexity grows linearly with N (the number of samples). However, real-world data exhibits diminishing returns: a dataset composed of a million identical images is barely more complex than one containing a single instance of that image.

We argue that data complexity should be *sub-additive*: the complexity of the union of two datasets should be at most the sum of their individual complexities, with equality only when the two sets are semantically disjoint. To achieve this, we move beyond counting samples and treat the dataset as a collection of points whose pairwise relationships are encoded by a pretrained generative model.

The key insight of this work is that score-based diffusion models [Song2021Score, Ho2020Denoising] provide a natural multi-scale representation of data geometry through the *probability flow ODE*. Given a learned score function $s(x, t) = \nabla_x \log p_t(x)$, each data point x defines a deterministic trajectory $z_x(t)$ that maps it from data space toward the noise prior. At intermediate time points, $z_x(t)$ encodes the semantic neighbourhood structure of x : the trajectory contracts toward the local cluster mean at small t and toward the global mean at large t , so the *path* a point takes through the ODE reveals its multi-scale identity. Recording these trajectories at L time points produces a *trajectory embedding*—a feature vector in $\mathbb{R}^{DL}$ that captures both fine-grained and coarse-grained data structure.

The Euclidean distance in this embedding space defines a *trajectory distance* $D(x_i, x_j)$ between data points. Two points are close in D if and only if they follow similar paths through the diffusion process at every time step—a strong condition that cannot be fooled by superficial similarity at any single scale. Applying a Gaussian kernel to the trajectory distance matrix produces a positive semi-definite kernel matrix K , and the *dataset complexity* $\mathcal{C}(\mathcal{X}) = \log \det(I + K)$ measures the total information content of the dataset.

This measure satisfies all properties one expects of a dataset complexity index: permutation invariance, bounded monotonicity (each new point contributes at most $\log 2$ bits), sub-additivity, submodularity (diminishing returns), and Lipschitz stability under data perturbation and score approximation error. Moreover, the measure has a clean information-theoretic interpretation: it counts the number of independent semantic “directions” the dataset occupies in trajectory space.

Contributions.

1. We define a dataset complexity measure $\mathcal{C}(\mathcal{X}) = \log \det(I + K)$ based on the probability flow ODE trajectory distance of a pretrained diffusion model (Section 3).

2. We prove that $\mathcal{C}$ satisfies permutation invariance, bounded monotonicity, sub-additivity, submodularity, and Lipschitz stability (Section 4).
3. We provide an information-theoretic characterisation of the asymptotic growth of $\mathcal{C}$ as dataset size increases (Section 3.9).
4. We discuss computational aspects, hyperparameter sensitivity, and the effect of score approximation error (Section 5).

2 Related Work

The quantification of data complexity has evolved from simple cardinality measures to sophisticated geometric and information-theoretic frameworks.

2.1 The manifold hypothesis and intrinsic dimension

The *Manifold Hypothesis* posits that high-dimensional data concentrates near low-dimensional submanifolds [Azeglio2025Inside, Kamkari2024Geometric]. Historically, *Intrinsic Dimension* (ID) has served as a proxy for complexity [Houle2015Inlierness]. Recent research has shifted toward *Local Intrinsic Dimension* (LID), which characterises the dimension of the submanifold in the immediate vicinity of a specific data point [Kamkari2024Geometric, Yu2026Local]. LID captures local factors of variation: complex data points exhibit higher local dimensionality.

While ID-based measures capture the “width” of the data manifold, they do not account for its global structure, curvature, or the redundancy between data points. Our measure addresses this gap by operating on the full pairwise distance structure rather than pointwise dimension estimates.

2.2 Score-based generative models as geometric probes

Score-based diffusion models learn the data distribution’s geometry through the Stein score function $s(x) = \nabla_x \log p(x)$ [Song2021Score, Ho2020Denoising]. The score encodes the manifold’s metric structure: Azeglio and Bernardo [Azeglio2025Inside] show that score functions can be used to derive a Riemannian metric tensor that stretches distances perpendicular to the manifold while preserving them along it, enabling the computation of geodesics that follow the manifold’s contours.

Riemannian curvature—the “shape” and “interconnectedness” of the manifold—determines the learning capacity required for generalisation [Sarkar2025Learning]. Negative curvature, characteristic of hyperbolic manifolds, results in exponential volume growth.

Our work exploits the score function—through the probability flow ODE—not to reconstruct the manifold but to define a multi-scale distance directly. This bypasses the need for explicit manifold reconstruction, Riemannian metric estimation, or eigenfunction computation.

2.3 Sub-additivity and dataset redundancy

As datasets scale, models encounter *sub-scaling*: performance gains decelerate due to high data density and redundancy [Chen2025SubScaling]. Information theory suggests that predictive information must grow sublinearly with dataset size [Anonymous2026TherML]. This motivates complexity measures $\mathcal{C}(\mathcal{X})$ satisfying $\mathcal{C}(\mathcal{A} \cup \mathcal{B}) \leq \mathcal{C}(\mathcal{A}) + \mathcal{C}(\mathcal{B})$ [Abin2026Subadditive].

The log-determinant functional $\log \det(I + K)$ is a classical tool in information theory and Gaussian process regression [CoverThomas2006]. It measures the mutual information between a latent function and noisy observations in the GP framework, and arises naturally as a submodular set function in experimental design [Krause2008Near]. Our contribution is to

combine this functional with a kernel derived from the score function of a generative model, grounding the abstract information-theoretic measure in the learned geometry of the data.

3 The Complexity Measure

3.1 The data model

Definition 3.1 (Data model). *Let p be a probability distribution on $\mathbb{R}^D$ supported on or near a low-dimensional set. A dataset $\mathcal{X} = \{x_1, \dots, x_N\} \subset \mathbb{R}^D$ is a finite collection of points drawn i.i.d. from p .*

We do *not* assume that the support of p is a smooth manifold, nor that we have access to the density p analytically. Instead, we assume access to a *pretrained score model* $s_\theta(x, t)$ that approximates the score function of the noisy marginals of p (see Section 3.2).

Why not assume a manifold? The manifold hypothesis is a useful modelling assumption but it complicates computation: it requires knowing the Riemannian metric, the eigenfunctions of the Laplace–Beltrami operator, or a graph-Laplacian approximation thereof. In practice, state-of-the-art generative models already encode the geometry of p implicitly through the score function. We exploit this learned geometry directly, bypassing the need to reconstruct the manifold.

3.2 Score-based diffusion models and the score function

The forward process. Given data $X_0 \sim p$, the *variance-exploding* (VE) forward process perturbs X_0 with Gaussian noise of increasing scale [Song2021Score]:

$$X_t = X_0 + \sigma(t) \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_D), \quad (1)$$

where $\sigma : [0, T] \rightarrow \mathbb{R}_{>0}$ is a strictly increasing noise schedule with $\sigma(0) \approx 0$ and $\sigma(T)$ large enough that $p_T \approx \mathcal{N}(0, \sigma(T)^2 I)$. The marginal density of X_t is the convolution:

$$p_t(x) = \int \frac{1}{(2\pi\sigma(t)^2)^{D/2}} \exp\left(-\frac{\|x - x_0\|^2}{2\sigma(t)^2}\right) p(x_0) dx_0. \quad (2)$$

The score function.

Definition 3.2 (Score function). *The score function at noise level t is:*

$$s(x, t) := \nabla_x \log p_t(x). \quad (3)$$

The score is the gradient of the log-density of the marginal at time t . It is the direction of steepest ascent of $\log p_t$ at the point x .

The MMSE denoising formula. The score has an explicit probabilistic interpretation as an optimal denoising direction.

Theorem 3.3 (Tweedie’s formula [Ho2020Denoising, Song2021Score]). *For the VE process (1):*

$$s(x, t) = \frac{\mathbb{E}[X_0 \mid X_t = x] - x}{\sigma(t)^2}. \quad (4)$$

Proof. Differentiating $\log p_t(x)$ from equation (2):

$$\begin{aligned}\nabla_x p_t(x) &= \int \frac{-(x - x_0)}{\sigma(t)^2} \cdot \frac{1}{(2\pi\sigma(t)^2)^{D/2}} \exp\left(-\frac{\|x - x_0\|^2}{2\sigma(t)^2}\right) p(x_0) dx_0 \\ &= -\frac{x}{\sigma(t)^2} p_t(x) + \frac{1}{\sigma(t)^2} \int x_0 p(x_0 | X_t = x) p_t(x) dx_0,\end{aligned}$$

where we used Bayes' rule to write $\mathcal{N}(x; x_0, \sigma^2)p(x_0) = p(x_0 | X_t = x)p_t(x)$. Dividing by $p_t(x)$:

$$s(x, t) = \frac{\nabla_x p_t(x)}{p_t(x)} = \frac{-x + \mathbb{E}[X_0 | X_t = x]}{\sigma(t)^2}.$$

□

Equivalently, the *denoised estimate*

$$\hat{x}_0(x, t) := \mathbb{E}[X_0 | X_t = x] = x + \sigma(t)^2 s(x, t) \quad (5)$$

is the minimum mean-squared error (MMSE) estimate of the clean data X_0 given the noisy observation $X_t = x$ [Ho2020Denoising].

Scale-dependent semantics. Equation (4) reveals the geometric content of the score at each noise level. Fix a data point $x_i \in \mathcal{X}$ and consider how $s(x_i, t)$ changes as $\sigma(t)$ varies:

- **Very small** $\sigma(t)$ (fine scale): the smoothed density p_t is nearly identical to p but with microscopic blur. The conditional expectation $\mathbb{E}[X_0 | X_t = x_i] \approx x_i$ (with high confidence, the clean datum is near x_i itself), so $s(x_i, t) \approx 0$. Scores of all clean data points are near zero; fine-scale differences are washed out.
- **Intermediate** $\sigma(t)$ (semantic scale): the Gaussian ball $\mathcal{N}(x_i; \cdot, \sigma^2)$ starts to overlap with the wider structure of p . The conditional expectation $\hat{x}_0(x_i, t)$ averages over training data within a neighbourhood of radius $\sim \sigma(t)$ of x_i , weighted by their density. Points in the same semantic cluster (same object class, same pose family) average to a similar $\hat{x}_0$; points in different clusters average to different $\hat{x}_0$. The score at this scale encodes *cluster membership*.
- **Very large** $\sigma(t)$ (global scale): the kernel $\mathcal{N}(x_i; \cdot, \sigma^2)$ is broad enough to see the whole distribution. For a distribution with mean μ , $\hat{x}_0(x_i, t) \approx \mu$ for all x_i , and $s(x_i, t) \approx (\mu - x_i)/\sigma(t)^2 \rightarrow 0$. Large-scale scores are small and convey only global position relative to the mean.

The *intermediate* scales carry the richest information. They are the scales at which the diffusion model resolves semantic distinctions between data points.

Practical computation. A pretrained diffusion model provides a network $s_\theta(x, t)$ trained to satisfy $s_\theta(x, t) \approx s(x, t)$ via *denoising score matching* [Ho2020Denoising, Song2021Score]:

$$\min_{\theta} \mathbb{E}_{t, X_0, \varepsilon} \left[\sigma(t)^2 \|s_\theta(X_0 + \sigma(t)\varepsilon, t) - s(X_0 + \sigma(t)\varepsilon, t)\|^2 \right]. \quad (6)$$

Equivalently, the model predicts the noise: defining $\hat{\varepsilon}_\theta(x, t) := -\sigma(t) s_\theta(x, t)$, equation (6) reduces to the standard DDPM noise-prediction objective $\mathbb{E}[\|\hat{\varepsilon}_\theta(X_t, t) - \varepsilon\|^2]$. Evaluating $s_\theta(x_i, t)$ amounts to one forward pass of the denoising network at the clean data point x_i with noise level $\sigma(t)$.

3.3 The probability flow ODE

The forward process (1) has an equivalent deterministic formulation: the *probability flow ODE* [Song2021Score] whose trajectories have the same marginal densities p_t as the stochastic process. For the variance-exploding schedule:

$$\frac{dz}{dt} = -\dot{\sigma}(t) \sigma(t) s(z, t), \quad z(0) = x, \quad (7)$$

where $\dot{\sigma}(t) = d\sigma/dt$. The solution $z_x(t)$ is the unique trajectory starting at the data point x at time $t = 0$. As t increases, $z_x(t)$ moves deterministically from data space toward the noise prior, passing through intermediate representations at each scale.

Trajectory semantics. The trajectory $z_x(t)$ has a rich semantic structure:

- **Small t :** $z_x(t) \approx x$; the trajectory has barely moved from the data point.
- **Intermediate t :** $z_x(t)$ contracts toward the local cluster mean. Points within the same cluster follow similar paths; points in different clusters diverge. The trajectory at this scale encodes *cluster membership*.
- **Large t :** $z_x(t)$ approaches the global mean of the distribution, regardless of the starting point.

Crucially, the trajectory $z_x(t)$ carries the full positional history of the point x through the ODE. Unlike the score evaluated at a fixed point, the trajectory does not lose cluster identity information for well-separated clusters: even when $s(x, t)$ depends only on the offset from the nearest cluster mean (so that cluster means cancel in score differences), the trajectory positions $z_x(t)$ retain the global position because $z(0) = x$.

3.4 The trajectory embedding

A single trajectory position $z_x(t_0)$ at a fixed time captures the data geometry at one scale only. To obtain a distance that is sensitive to structure at *all* semantically relevant scales simultaneously, we integrate trajectory divergence over time with a weighting that emphasises the most informative scales.

Definition 3.4 (Trajectory distance, continuous form). *Let $w : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ be a probability density on $[0, 1]$. The weighted trajectory distance between two data points is:*

$$D(x_i, x_j)^2 := \int_0^1 \|z_{x_i}(t) - z_{x_j}(t)\|^2 w(t) dt = \mathbb{E}_{t \sim w} [\|z_{x_i}(t) - z_{x_j}(t)\|^2]. \quad (8)$$

The trajectory distance is the *expected squared divergence* between two ODE trajectories, where the expectation is over a random time drawn from the weighting distribution w .

Beta weighting. A natural two-parameter family for w is the Beta distribution:

$$w(t) = \text{Beta}(t; \alpha, \beta) = \frac{t^{\alpha-1}(1-t)^{\beta-1}}{B(\alpha, \beta)}, \quad \alpha, \beta > 0, \quad (9)$$

where $B(\alpha, \beta)$ is the Beta function. The two parameters provide intuitive control over which time scales dominate:

- $\alpha = \beta = 1$: uniform weighting (all scales contribute equally).
- $\alpha > 1, \beta > 1$: emphasis on intermediate t (the regime where within-cluster noise has been reduced but between-cluster separation persists — the semantic sweet spot).

- $\alpha < 1, \beta > 1$: emphasis on small t (fine-grained positional structure).
- $\alpha > 1, \beta < 1$: emphasis on large t (coarse global structure).

The mode of $\text{Beta}(\alpha, \beta)$ is $(\alpha - 1)/(\alpha + \beta - 2)$ for $\alpha, \beta > 1$, giving direct control over the centre of the weighting, while the concentration $\alpha + \beta$ controls the width.

Definition 3.5 (Trajectory embedding). *In practice, the integral (??) is approximated by quadrature. Let $0 < t_1 < t_2 < \dots < t_L < 1$ be L quadrature nodes and let $w_k = w(t_k)/\sum_{l=1}^L w(t_l)$ be the normalised quadrature weights. The trajectory embedding is the map $\Phi : \mathbb{R}^D \rightarrow \mathbb{R}^{DL}$ defined by:*

$$\Phi(x) := (\sqrt{w_1} z_x(t_1), \sqrt{w_2} z_x(t_2), \dots, \sqrt{w_L} z_x(t_L)), \quad (10)$$

where $z_x(t)$ solves the probability flow ODE (7) with initial condition $z_x(0) = x$. Each block $\sqrt{w_k} z_x(t_k) \in \mathbb{R}^D$ contributes one time scale; the full vector $\Phi(x) \in \mathbb{R}^{DL}$ is the concatenation.

The squared distance in the embedding space approximates the integral (??):

$$\|\Phi(x_i) - \Phi(x_j)\|^2 = \sum_{k=1}^L w_k \|z_{x_i}(t_k) - z_{x_j}(t_k)\|^2 \approx \mathbb{E}_{t \sim w} [\|z_{x_i}(t) - z_{x_j}(t)\|^2]. \quad (11)$$

Interpretation as a fingerprint. $\Phi(x)$ is a *multi-scale fingerprint* of the data point x : a weighted record of how the data point traverses the probability flow ODE at L different time points. Each block $\sqrt{w_k} z_x(t_k)$ encodes where x has been transported to by time t_k —i.e., the intermediate representation of x at scale $\sigma(t_k)$. Points that the generative model processes identically at all scales follow identical trajectories; points that the model distinguishes at any scale follow diverging trajectories.

Remark 3.6 (Monte Carlo estimation). *An alternative to a fixed quadrature grid is to draw the evaluation times as i.i.d. samples $t_1, \dots, t_L \sim \text{Beta}(\alpha, \beta)$. The embedding (8) with uniform weights $w_k = 1/L$ then yields an unbiased Monte Carlo estimate of (??). The CND and PSD properties hold for any fixed realisation of $\{t_k\}$, since the embedding is Euclidean in $\mathbb{R}^{DL}$ regardless of how the time points are chosen.*

Remark 3.7 (Jensen lower bound). *Since $\|\cdot\|^2$ is convex, Jensen’s inequality gives:*

$$D(x_i, x_j)^2 = \mathbb{E}_t [\|z_{x_i}(t) - z_{x_j}(t)\|^2] \geq \|\mathbb{E}_t [z_{x_i}(t) - z_{x_j}(t)]\|^2 = \|\bar{z}_{x_i} - \bar{z}_{x_j}\|^2,$$

where $\bar{z}_x = \mathbb{E}_{t \sim w} [z_x(t)]$ is the mean trajectory position. This provides a cheap lower bound on D^2 from a single D -dimensional vector per point, useful for fast approximate screening of candidate points.

Why trajectories instead of scores. A simpler alternative is the *score embedding* $\Phi^S(x) = (s(x, t_1), \dots, s(x, t_L))$, which concatenates scores evaluated at the fixed data point x . While this also yields a Euclidean distance in $\mathbb{R}^{DL}$ with the same CND and PSD properties, it suffers from a fundamental limitation: for well-separated clusters, the score at a data point x near cluster mean μ_k is approximately $s(x, t) \approx -(x - \mu_k)/(\sigma_k^2 + \sigma(t)^2)$. The score difference between two points from different clusters is:

$$s(x_1, t) - s(x_2, t) \approx \frac{(x_2 - \mu_j) - (x_1 - \mu_i)}{\sigma_k^2 + \sigma(t)^2} = \frac{\varepsilon_2 - \varepsilon_1}{\sigma_k^2 + \sigma(t)^2},$$

where $\varepsilon_i = x_i - \mu_i$ are the offsets from the respective cluster means. The cluster means μ_i, μ_j cancel completely, making the score distance blind to cluster identity.

The trajectory embedding does not suffer from this cancellation. At any time $t > 0$, the trajectory position $z_x(t)$ depends on the *absolute* initial position x (since $z(0) = x$), not merely the offset from the nearest mean. Thus $z_{x_1}(t) - z_{x_2}(t)$ retains the cluster separation $\mu_i - \mu_j$ as a dominant contribution at intermediate times.

Remark 3.8 (Connection to the denoised estimate). *By Tweedie’s formula (4), the MMSE denoised estimate $\hat{x}_0(x, t) = x + \sigma(t)^2 s(x, t)$ satisfies the same cancellation as the score itself: for well-separated clusters, $\hat{x}_0(x_1, t) - \hat{x}_0(x_2, t)$ also loses cluster identity. This is because the denoised estimate is a pointwise function of x at a fixed noise level, just as the score is. The trajectory embedding avoids this by integrating the score field along the flow, so that the embedding at time t depends on the full path from $z(0) = x$ to $z(t)$.*

3.5 The trajectory distance

The trajectory distance (Definition ??) is defined continuously as an expectation; in practice it is computed via the finite-dimensional embedding (Definition 3.4).

Definition 3.9 (Trajectory distance, discrete form). *The trajectory distance between two data points is the Euclidean distance between their trajectory embeddings:*

$$D(x_i, x_j) := \|\Phi(x_i) - \Phi(x_j)\|_{\mathbb{R}^{DL}} = \left(\sum_{k=1}^L w_k \|z_{x_i}(t_k) - z_{x_j}(t_k)\|^2 \right)^{1/2}. \quad (12)$$

Metric properties. Since $D(x_i, x_j) = \|\mathbf{a}_i - \mathbf{a}_j\|$ for $\mathbf{a}_i = \Phi(x_i) \in \mathbb{R}^{DL}$, the function D inherits all metric axioms from the Euclidean norm:

1. *Non-negativity:* $D(x_i, x_j) \geq 0$, with equality iff $\Phi(x_i) = \Phi(x_j)$.
2. *Symmetry:* $D(x_i, x_j) = D(x_j, x_i)$.
3. *Triangle inequality:* for any three points, $D(x_i, x_j) \leq D(x_i, x_k) + D(x_k, x_j)$.

The triangle inequality follows directly from $\|\mathbf{a} - \mathbf{b}\| \leq \|\mathbf{a} - \mathbf{c}\| + \|\mathbf{c} - \mathbf{b}\|$ for any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^{DL}$.

In general D is a *pseudometric*: $D(x_i, x_j) = 0$ implies $\Phi(x_i) = \Phi(x_j)$, not necessarily $x_i = x_j$. Two distinct data points could, in principle, follow identical ODE trajectories. For generic data and a well-trained model, this does not occur.

Per-scale decomposition. Writing D^2 as a weighted sum over time points:

$$D(x_i, x_j)^2 = \sum_{k=1}^L w_k \underbrace{\|z_{x_i}(t_k) - z_{x_j}(t_k)\|^2}_{=: \delta_k(x_i, x_j)}, \quad (13)$$

each term δ_k is the squared trajectory divergence at time t_k . When w is a Beta density with $\alpha, \beta > 1$, the intermediate time points receive the highest weight — precisely the regime where within-cluster noise has been reduced by the ODE contraction but between-cluster separation persists.

Semantic interpretation. Two points x_i, x_j are *close* in D if and only if their probability flow trajectories remain close at every time point weighted by w . This is a strong condition:

- It fails at small t when x_i, x_j are far apart in data space (trajectories start at different positions, $z(0) = x$).
- It fails at intermediate t when x_i, x_j belong to different semantic clusters (trajectories contract toward different cluster means).
- It fails at large t only if trajectories have not yet converged to the global mean.

- All conditions must hold simultaneously across all weighted time points for $D(x_i, x_j)$ to be small.

Unlike the score embedding, which evaluates the score at the fixed position x and therefore loses cluster identity for well-separated clusters (Section 3.4), the trajectory embedding preserves cluster identity because the ODE trajectory carries the initial position $z(0) = x$ through the flow.

3.6 The kernel

Definition 3.10 (Trajectory kernel). *For parameters $\lambda > 0$ (bandwidth), time points $0 < t_1 < \dots < t_L$, and weights $w_1, \dots, w_L > 0$, define the trajectory kernel:*

$$K_{ij} = K_\lambda(x_i, x_j) := \exp(-\lambda D(x_i, x_j)^2) = \exp\left(-\lambda \sum_{k=1}^L w_k \|z_{x_i}(t_k) - z_{x_j}(t_k)\|^2\right). \quad (14)$$

Since $\Phi : \mathbb{R}^D \rightarrow \mathbb{R}^{DL}$ and $D(x_i, x_j) = \|\Phi(x_i) - \Phi(x_j)\|_{\mathbb{R}^{DL}}$, the kernel (11) is precisely the standard Gaussian (RBF) kernel applied in the feature space $\mathbb{R}^{DL}$:

$$K_{ij} = \exp(-\lambda \|\Phi(x_i) - \Phi(x_j)\|^2). \quad (15)$$

Gaussian kernels on finite-dimensional Euclidean spaces are always PSD (proved formally in Corollary 4.7).

Role of the parameters.

- **Bandwidth λ :** controls the width of the similarity function. Large λ makes only very nearby points (in trajectory space) similar; small λ makes the similarity decay slowly.
- **Time grid $\{t_1, \dots, t_L\}$ and weights $\{w_1, \dots, w_L\}$:** determine which time scales contribute to the distance and how much. Emphasising intermediate times focuses the kernel on semantic cluster structure; emphasising early times adds sensitivity to fine-grained variation.

3.7 The complexity measure

Definition 3.11 (Dataset complexity). *Let $K \in \mathbb{R}^{N \times N}$ be the kernel matrix with entries (11). The complexity of $\mathcal{X}$ is:*

$$\mathcal{C}(\mathcal{X}) = \log \det(I + K). \quad (16)$$

Spectral expansion. Let $\mu_1 \geq \mu_2 \geq \dots \geq \mu_N \geq 0$ be the eigenvalues of K (non-negative since $K \succeq 0$). Then:

$$\mathcal{C}(\mathcal{X}) = \sum_{k=1}^N \log(1 + \mu_k). \quad (17)$$

Each term $\log(1 + \mu_k)$ is the contribution of the k -th independent direction in trajectory space to the total complexity.

Diminishing returns. The function $\mu \mapsto \log(1 + \mu)$ is strictly concave. Consequently:

- A large eigenvalue $\mu_k \gg 1$ (a dominant, strongly coherent direction in trajectory space) contributes $\approx \log \mu_k$ — significant but growing slowly.
- A small eigenvalue $\mu_k \ll 1$ (a weak direction, arising when a new point is nearly collinear with existing points in $\mathbb{R}^{DL}$) contributes $\approx \mu_k$.

- An eigenvalue $\mu_k \approx 0$ (a redundant point whose multi-scale trajectory is nearly identical to a combination of existing ones) contributes essentially zero.

Adding a point whose trajectory is already well represented by the existing dataset contributes nearly zero to $\mathcal{C}$. Adding a point with a genuinely novel trajectory always increases $\mathcal{C}$.

3.8 Information-theoretic interpretation

Gaussian process mutual information. Consider a Gaussian process $f \sim \mathcal{GP}(0, K)$ observed at $\mathcal{X}$ with unit i.i.d. noise: $y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 1)$. The vector $\mathbf{f} = (f(x_i))_i \sim \mathcal{N}(0, K)$ and $\mathbf{y} \mid \mathbf{f} \sim \mathcal{N}(\mathbf{f}, I)$. By the Gaussian mutual information formula [CoverThomas2006]:

$$I(\mathbf{f}; \mathbf{y}) = \frac{1}{2} \log \det(I + K) = \frac{1}{2} \mathcal{C}(\mathcal{X}). \quad (18)$$

$\mathcal{C}(\mathcal{X})$ is twice the mutual information between the latent GP values at $\mathcal{X}$ and noisy observations: the number of bits the dataset can carry about the latent function.

In this context, the kernel K plays the role of the prior covariance of the latent function. The trajectory kernel $K_{ij} = \exp(-\lambda D^2)$ assigns high prior covariance to pairs of points that follow similar paths through the diffusion process. $\mathcal{C}(\mathcal{X})$ then counts how many independent semantic “directions” the dataset occupies.

Effective dimensionality. The quantity $\mathcal{C}(\mathcal{X}) = \sum_k \log(1 + \mu_k)$ is related to the *effective rank* of K : it counts the number of eigenvalues of K that are substantially positive, weighted by the amount of information each contributes. Unlike the algebraic rank (which equals N for generic data and is discontinuous), $\mathcal{C}$ is smooth and saturates naturally as new points become redundant in trajectory space.

Why not other functionals?

- $\text{tr}(K) = \sum_k \mu_k$: adding a duplicate point (same trajectory as an existing one, $\mu \approx 1$) increases the trace by ~ 1 regardless of how many identical points already exist. Not sub-additive in general.
- $\log \det(K) = \sum_k \log \mu_k$: undefined when K is singular (generic for $N > DL$) and not monotone.
- $\lambda_{\max}(K)$: ignores all but the largest eigenvalue; cannot distinguish a dataset with one dominant semantic direction from one with many independent directions.

The log-det of $I + K$ avoids all three issues and is the canonical information-theoretic measure for Gaussian processes.

3.9 Asymptotic behaviour

Case 1: N identical points. If $x_1 = \dots = x_N = x_0$, then $\Phi(x_i) = \Phi(x_j)$ for all i, j , so $D(x_i, x_j) = 0$ and $K_{ij} = 1$ for all i, j . The kernel matrix is $K = \mathbf{1}\mathbf{1}^\top$ with eigenvalues N and 0 ($N - 1$ times). By the matrix determinant lemma:

$$\mathcal{C}(\{x_0, \dots, x_0\}) = \log(1 + N) \approx \log N. \quad (19)$$

Complexity grows *logarithmically*: each additional copy contributes $\approx 1/N$ bits, since it is fully predictable from existing copies.

Case 2: N maximally diverse points. If the points are so well-separated in trajectory space that $D(x_i, x_j) \rightarrow \infty$ for all $i \neq j$ (equivalently $K_{ij} \rightarrow 0$ for $i \neq j$), then $K \rightarrow I$ and:

$$\mathcal{C}(\mathcal{X}) \rightarrow N \log 2. \quad (20)$$

Complexity grows *linearly*: each additional maximally diverse point contributes exactly $\log 2$ bits of new semantic information.

Interpolation: M clusters. Suppose $\mathcal{X}$ consists of M well-separated clusters, each containing $m = N/M$ points with identical trajectories. Within each cluster $K_{ij} = 1$; between clusters $K_{ij} = 0$. The kernel matrix is block-diagonal with M blocks of size m . Each block contributes $\log(1 + m)$, giving:

$$\mathcal{C}(\mathcal{X}) = M \log(1 + m) = M \log\left(\frac{N}{M} + 1\right) \approx M \log \frac{N}{M}. \quad (21)$$

The complexity is determined by the number of independent semantic clusters M , not by the total number of points N . Doubling N by duplicating each cluster member changes $m \rightarrow 2m$ but does not change M , so $\mathcal{C}$ increases only logarithmically in N . Adding a genuinely new cluster (increasing M) adds $\log(N/M + 1)$ — a much larger gain.

Information-theoretic interpretation of growth. The growth rate of $\mathcal{C}$ has a clean interpretation. Even for maximally redundant data (N identical points), $\mathcal{C}$ grows as $\log(1 + N) \approx \log N$. This is not a defect: it is the *information-theoretic floor*—the $\log N$ bits required to encode the dataset size itself. Any dataset of N objects requires at least $\log N$ bits to describe, regardless of its content.

The meaningful quantity is the *growth rate*, which distinguishes:

- $\Theta(\log N)$: trivially redundant data (all information is in the count).
- $\Theta(M \log N)$: data with M independent semantic directions (structure plus count).
- $\Theta(N)$: maximally diverse data (each point carries novel information).

The growth rate exponent is the intrinsic quantity characterising the distribution’s complexity. $\mathcal{C}$ measures *dataset* complexity (which naturally grows with N), not *distribution* complexity (which would converge).

4 Properties

Throughout this section, K denotes the $N \times N$ kernel matrix with entries $K_{ij} = \exp(-\lambda D(x_i, x_j)^2)$ for fixed $\lambda > 0$ and time grid $\{t_k\}$. The key object is the trajectory embedding $\Phi : \mathbb{R}^D \rightarrow \mathbb{R}^{DL}$ from Definition 3.4. We write $\mathbf{a}_i = \Phi(x_i) \in \mathbb{R}^{DL}$ for short. $k_x \in \mathbb{R}^N$ denotes the vector $(K(x, x_i))_{i=1}^N$ of similarities between a new point x and the existing dataset $\mathcal{X}$.

All proofs below depend only on the fact that Φ maps into a Euclidean space $\mathbb{R}^{DL}$ and the kernel is a Gaussian RBF applied to the resulting Euclidean distances. They hold verbatim for any embedding with this structure, including the trajectory embedding (8).

4.1 Permutation invariance

Proposition 4.1 (Permutation invariance). $\mathcal{C}(\mathcal{X})$ does not depend on the ordering of points.

Proof. Any reordering corresponds to conjugating K by a permutation matrix P (orthogonal, $P^\top P = I$):

$$\det(I + PKP^\top) = \det(P(I + K)P^\top) = \det(P) \det(I + K) \det(P^\top) = \det(I + K).$$

□

4.2 The trajectory distance squared is conditionally negative definite

Definition 4.2 (Conditionally negative definite [BergChristensenRessel1984]). A symmetric function $\psi : X \times X \rightarrow \mathbb{R}$ with $\psi(x, x) = 0$ is conditionally negative definite (CND) if for all $n \geq 1$, all $x_1, \dots, x_n \in X$, and all $c_1, \dots, c_n \in \mathbb{R}$ with $\sum_{i=1}^n c_i = 0$:

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j \psi(x_i, x_j) \leq 0.$$

Theorem 4.3 (D^2 is CND). The function $(x_i, x_j) \mapsto D(x_i, x_j)^2$ is conditionally negative definite.

Proof. For any $c_1, \dots, c_n$ with $\sum_i c_i = 0$:

$$\begin{aligned} \sum_{i,j} c_i c_j D(x_i, x_j)^2 &= \sum_{i,j} c_i c_j \|\mathbf{a}_i - \mathbf{a}_j\|^2 \\ &= \sum_{i,j} c_i c_j (\|\mathbf{a}_i\|^2 - 2\langle \mathbf{a}_i, \mathbf{a}_j \rangle + \|\mathbf{a}_j\|^2) \\ &= \underbrace{\left(\sum_j c_j\right)}_{=0} \sum_i c_i \|\mathbf{a}_i\|^2 + \underbrace{\left(\sum_i c_i\right)}_{=0} \sum_j c_j \|\mathbf{a}_j\|^2 - 2 \left\langle \sum_i c_i \mathbf{a}_i, \sum_j c_j \mathbf{a}_j \right\rangle \\ &= -2 \left\| \sum_i c_i \mathbf{a}_i \right\|^2 \leq 0. \end{aligned}$$

□

Remark 4.4. This proof uses only that $D(x_i, x_j) = \|\mathbf{a}_i - \mathbf{a}_j\|$ is the Euclidean distance in some Hilbert space. The specific form of $\mathbf{a}_i = \Phi(x_i)$ (scores concatenated from a generative model) is irrelevant to the CND property. The same argument works for any feature map $\Phi : \mathbb{R}^D \rightarrow H$ into any Hilbert space H .

4.3 The trajectory distance is conditionally negative definite

Theorem 4.5 (D is CND). The function $(x_i, x_j) \mapsto D(x_i, x_j)$ is conditionally negative definite.

Proof. By Theorem 4.3, D^2 is CND. The function $s \mapsto \sqrt{s}$ is a Bernstein function: it is non-negative on $(0, \infty)$ and has a completely monotone derivative $(\sqrt{s})' = \frac{1}{2}s^{-1/2}$ (completely monotone as a Laplace transform of the positive measure $\frac{1}{2\sqrt{\pi}}u^{-3/2}du$). A composition of a CND function with a Bernstein function is again CND [BergChristensenRessel1984]. Therefore $D = \sqrt{D^2}$ is CND.

Alternative: In any Hilbert space H , the function $(x, y) \mapsto \|x - y\|_H^\alpha$ is CND for all $\alpha \in (0, 2]$ [BergChristensenRessel1984]. Taking $\alpha = 1$ and $H = \mathbb{R}^{DL}$ with the Euclidean norm gives the result directly. □

4.4 Schoenberg's theorem

Theorem 4.6 (Schoenberg [Schoenberg1938Metric, BergChristensenRessel1984]). Let $\psi : X \times X \rightarrow \mathbb{R}$ be a continuous symmetric function with $\psi(x, x) = 0$. Then ψ is CND if and only if $e^{-\lambda\psi(x,y)}$ is PSD for all $\lambda > 0$.

A complete proof is in [BergChristensenRessel1984]. Geometrically [Feragen2015Geodesic], a metric space (X, d) embeds isometrically into a Hilbert space if and only if d^2 is CND, which by Schoenberg is equivalent to $e^{-\lambda d^2}$ being PSD for all $\lambda > 0$. Since D is the Euclidean distance in $\mathbb{R}^{DL}$, the embedding is Φ itself, and D^2 is CND (Theorem 4.3) by direct construction.

4.5 The proposed kernel is positive semi-definite

Corollary 4.7 (Gaussian trajectory kernel is PSD). *For all $\lambda > 0$, the kernel $K_{ij} = \exp(-\lambda D(x_i, x_j)^2)$ is PSD.*

Proof. D^2 is CND by Theorem 4.3. Schoenberg’s theorem (Theorem 4.6) gives $\exp(-\lambda D^2)$ PSD for all $\lambda > 0$. Alternatively: $K_{ij} = \exp(-\lambda \|\mathbf{a}_i - \mathbf{a}_j\|^2)$ is the standard Gaussian RBF kernel applied in the Euclidean space $\mathbb{R}^{DL}$, which is PSD by the standard argument (Mercer’s theorem or Bochner’s theorem). $\square$

Corollary 4.8 (Laplacian trajectory kernel is PSD). *For all $\lambda > 0$, the kernel $K_{ij}^L = \exp(-\lambda D(x_i, x_j))$ is PSD.*

Proof. D is CND by Theorem 4.5. Apply Schoenberg’s theorem. $\square$

Remark 4.9. *The Laplacian kernel (Corollary 4.8) may be preferable when the distribution of trajectory distances is heavy-tailed: the Laplacian kernel decays more slowly than the Gaussian, giving non-negligible similarity to moderately distant points. Both kernels have identical theoretical properties; the remainder of this document works with the Gaussian kernel (Corollary 4.7).*

4.6 Bounded monotonicity

Theorem 4.10 (Bounded monotonicity). *For any dataset $\mathcal{X}$ and any point $x \in \mathbb{R}^D$:*

$$0 < \mathcal{C}(\mathcal{X} \cup \{x\}) - \mathcal{C}(\mathcal{X}) \leq \log 2. \quad (22)$$

The upper bound $\log 2$ is achieved in the limit $D(x, x_i) \rightarrow \infty$ for all $x_i \in \mathcal{X}$ (a maximally novel point contributes $\log 2$ bits). The gain approaches 0 as x becomes redundant in a growing dataset, but is always strictly positive.

Proof. Write the enlarged kernel matrix in block form:

$$I + K_{\mathcal{X} \cup \{x\}} = \begin{pmatrix} I + K_{\mathcal{X}} & k_x \\ k_x^\top & 1 + K(x, x) \end{pmatrix},$$

where $k_x = (K(x, x_i))_{i=1}^N \in \mathbb{R}^N$ and $K(x, x) = \exp(-\lambda D(x, x)^2) = \exp(0) = 1$. By the Schur complement formula:

$$\det(I + K_{\mathcal{X} \cup \{x\}}) = \det(I + K_{\mathcal{X}}) \cdot S, \quad S := 1 + \sigma^2(x | \mathcal{X}), \quad (23)$$

where $\sigma^2(x | \mathcal{X}) := K(x, x) - k_x^\top (I + K_{\mathcal{X}})^{-1} k_x$ is the Schur complement of the $(1, 1)$ -block minus 1. The inverse exists because $I + K_{\mathcal{X}} \succ 0$ ($K_{\mathcal{X}} \succeq 0$). The marginal complexity gain is:

$$\mathcal{C}(\mathcal{X} \cup \{x\}) - \mathcal{C}(\mathcal{X}) = \log S = \log(1 + \sigma^2(x | \mathcal{X})).$$

Upper bound. Since $k_x^\top (I + K_{\mathcal{X}})^{-1} k_x \geq 0$ (quadratic form with PSD matrix), we have $\sigma^2(x | \mathcal{X}) \leq K(x, x) = 1$, and therefore $S \leq 2$. Equality holds when $k_x = \mathbf{0}$, i.e. $K(x, x_i) = 0$ for all i , which requires $D(x, x_i) \rightarrow \infty$ for all i .

Lower bound. We show $\sigma^2(x | \mathcal{X}) > 0$, hence $S > 1$ and $\log S > 0$. Since $K_{\mathcal{X}} \succeq 0$, we have $I + K_{\mathcal{X}} \succeq I$, so $(I + K_{\mathcal{X}})^{-1} \preceq I$. Therefore:

$$k_x^\top (I + K_{\mathcal{X}})^{-1} k_x \leq k_x^\top k_x = \sum_{i=1}^N K(x, x_i)^2.$$

If $k_x \neq \mathbf{0}$, the inequality is strict (since $(I + K_{\mathcal{X}})^{-1} \prec I$ when $K_{\mathcal{X}} \neq 0$), giving $\sigma^2 = 1 - k_x^\top (I + K_{\mathcal{X}})^{-1} k_x > 0$. If $k_x = \mathbf{0}$, then $\sigma^2 = K(x, x) = 1 > 0$. In both cases $S > 1$. $\square$

Remark 4.11 (Per-point information budget). *Theorem 4.10 establishes that each data point contributes between 0 and $\log 2$ bits to $\mathcal{C}$. For N data points:*

$$0 < \mathcal{C}(\mathcal{X}) \leq N \log 2. \quad (24)$$

The lower bound $\log(1 + N)$ (from N identical points) and upper bound $N \log 2$ (from N maximally diverse points) match the asymptotic analysis of Section 3.9.

Remark 4.12 (Kernel fixedness and monotonicity). *A crucial distinction: monotonicity holds because the kernel $K(x_i, x_j) = \exp(-\lambda D(x_i, x_j)^2)$ is a fixed function of the data points, not of the current dataset $\mathcal{X}$. Adding a new point x to $\mathcal{X}$ does not change the existing entries $K(x_i, x_j)$ for $i, j \in \mathcal{X}$. This is precisely why the trajectory-based kernel is preferable to the Coifman–Lafon graph-Laplacian kernel (in which adding a point shifts all node degrees and changes every entry of the kernel matrix).*

4.7 Sub-additivity

Theorem 4.13 (Sub-additivity). *For any two disjoint datasets $\mathcal{A}$ and $\mathcal{B}$:*

$$\mathcal{C}(\mathcal{A} \cup \mathcal{B}) \leq \mathcal{C}(\mathcal{A}) + \mathcal{C}(\mathcal{B}).$$

Proof. Write the kernel matrix of $\mathcal{A} \cup \mathcal{B}$ in block form:

$$I + K_{\mathcal{A} \cup \mathcal{B}} = \begin{pmatrix} I + K_{\mathcal{A}} & K_{\mathcal{A}\mathcal{B}} \\ K_{\mathcal{A}\mathcal{B}}^\top & I + K_{\mathcal{B}} \end{pmatrix},$$

where $(K_{\mathcal{A}\mathcal{B}})_{ij} = K(a_i, b_j)$. This matrix is PSD. We apply *Fischer’s inequality*:

Fischer’s inequality. For any PSD block matrix $M = \begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}$ with $A, B \succ 0$: $\det M \leq \det A \cdot \det B$ [HornJohnson2013].

Proof of Fischer’s inequality. Since $A \succ 0$, the Schur complement $S_A = B - C^\top A^{-1}C$ satisfies $S_A \succeq 0$ (from $M \succeq 0$).

$$\det M = \det A \cdot \det(B - C^\top A^{-1}C).$$

Let $Y = B^{-1/2}C^\top A^{-1}CB^{-1/2} \succeq 0$. Then $B - C^\top A^{-1}C = B^{1/2}(I - Y)B^{1/2}$, so $\det(B - C^\top A^{-1}C) = \det B \cdot \det(I - Y)$. Since $S_A \succeq 0$ implies $I \succeq Y$, all eigenvalues ξ_k of Y lie in $[0, 1]$, giving $\det(I - Y) = \prod_k (1 - \xi_k) \leq 1$. Therefore $\det M \leq \det A \cdot \det B$.

Applying this to $M = I + K_{\mathcal{A} \cup \mathcal{B}}$, $A = I + K_{\mathcal{A}}$, $B = I + K_{\mathcal{B}}$:

$$\det(I + K_{\mathcal{A} \cup \mathcal{B}}) \leq \det(I + K_{\mathcal{A}}) \cdot \det(I + K_{\mathcal{B}}).$$

Taking logarithms gives the result. □

Remark 4.14 (The gap measures shared semantic content). *Equality in Fischer’s inequality holds when $C = 0$, i.e. when every cross-pair (a_i, b_j) satisfies $K(a_i, b_j) = 0$. Since $K(a, b) = \exp(-\lambda D(a, b)^2) > 0$ whenever the trajectories of a and b are at finite distance, strict sub-additivity always holds for finite, non-degenerate datasets. The gap*

$$\mathcal{C}(\mathcal{A}) + \mathcal{C}(\mathcal{B}) - \mathcal{C}(\mathcal{A} \cup \mathcal{B}) \geq 0$$

measures how much semantic content $\mathcal{A}$ and $\mathcal{B}$ share: it is large when many cross-pairs have small trajectory distance (similar semantic content), and small when $\mathcal{A}$ and $\mathcal{B}$ are semantically disjoint.

4.8 Submodularity

Theorem 4.15 (Submodularity / diminishing returns). *For any $\mathcal{X} \subseteq \mathcal{X}'$ and any point $x \notin \mathcal{X}'$:*

$$\mathcal{C}(\mathcal{X} \cup \{x\}) - \mathcal{C}(\mathcal{X}) \geq \mathcal{C}(\mathcal{X}' \cup \{x\}) - \mathcal{C}(\mathcal{X}').$$

Each additional data point contributes less complexity when more data is already present.

Proof. By the Schur complement formula (20), the marginal gain from adding x to a dataset $\mathcal{Z}$ is:

$$\mathcal{C}(\mathcal{Z} \cup \{x\}) - \mathcal{C}(\mathcal{Z}) = \log(1 + \sigma^2(x | \mathcal{Z})),$$

where $\sigma^2(x | \mathcal{Z}) = K(x, x) - k_{x, \mathcal{Z}}^\top (I + K_{\mathcal{Z}})^{-1} k_{x, \mathcal{Z}}$.

Since $\log(1 + \cdot)$ is non-decreasing, it suffices to show $\sigma^2(x | \mathcal{X}) \geq \sigma^2(x | \mathcal{X}')$ for $\mathcal{X} \subseteq \mathcal{X}'$. Write $\mathcal{X}' = \mathcal{X} \cup \mathcal{Y}$. By the sequential conditioning formula for Gaussians:

$$\sigma^2(x | \mathcal{X}') = \sigma^2(x | \mathcal{X}) - k_{x, \mathcal{Y} | \mathcal{X}}^\top (I + K_{\mathcal{Y} | \mathcal{X}})^{-1} k_{x, \mathcal{Y} | \mathcal{X}},$$

where the subscript $|\mathcal{X}$ denotes residuals after conditioning. The second term is non-negative (quadratic form with PSD matrix), so $\sigma^2(x | \mathcal{X}') \leq \sigma^2(x | \mathcal{X})$. Therefore the marginal gain is non-increasing in $\mathcal{X}$, which is submodularity. $\square$

Remark 4.16 (Greedy data selection). *Submodularity implies that the greedy algorithm for maximising $\mathcal{C}$ subject to a cardinality constraint — at each step, add the point of maximum marginal gain — achieves a $(1 - 1/e)$ -approximation to the optimal subset of any fixed size. This makes $\mathcal{C}$ useful not only as a complexity measure but as an objective for data selection: greedily selecting the n points that maximise complexity gives a near-optimal diverse subset.*

4.9 Lipschitz continuity and stability

Both data perturbation and score approximation error affect $\mathcal{C}$ through perturbation of the trajectory embedding $\Phi(x_i)$. We prove a unified bound that covers both cases.

Lemma 4.17 (Master perturbation bound). *Let K, K' be two $N \times N$ PSD kernel matrices obtained from embeddings $\mathbf{a}_i, \mathbf{a}'_i$ with $\eta := \max_i \|\mathbf{a}'_i - \mathbf{a}_i\|$. Let $D_{\max} = \max_{i,j} \|\mathbf{a}_i - \mathbf{a}_j\|$. Then:*

$$|\mathcal{C}'(\mathcal{X}) - \mathcal{C}(\mathcal{X})| \leq 4N^{3/2} \lambda \eta (D_{\max} + \eta). \quad (25)$$

Proof. **Step 1** (distance perturbation). Writing $\mathbf{a}'_i = \mathbf{a}_i + \boldsymbol{\epsilon}_i$ with $\|\boldsymbol{\epsilon}_i\| \leq \eta$:

$$\begin{aligned} |D'(x_i, x_j)^2 - D(x_i, x_j)^2| &= \left| \|\mathbf{a}'_i - \mathbf{a}'_j\|^2 - \|\mathbf{a}_i - \mathbf{a}_j\|^2 \right| \\ &= \left| 2\langle \mathbf{a}_i - \mathbf{a}_j, \boldsymbol{\epsilon}_i - \boldsymbol{\epsilon}_j \rangle + \|\boldsymbol{\epsilon}_i - \boldsymbol{\epsilon}_j\|^2 \right| \\ &\leq 2D_{\max} \cdot 2\eta + (2\eta)^2 = 4\eta(D_{\max} + \eta). \end{aligned}$$

Step 2 (kernel perturbation). Since $|e^{-a} - e^{-b}| \leq |a - b|$ for $a, b \geq 0$:

$$|K'_{ij} - K_{ij}| \leq \lambda |D'(x_i, x_j)^2 - D(x_i, x_j)^2| \leq 4\lambda \eta (D_{\max} + \eta) =: \alpha.$$

Step 3 (Frobenius norm). Since the bound α applies to each of the N^2 entries: $\|K' - K\|_F \leq N\alpha$.

Step 4 (log-det perturbation). Define $g(t) = \log \det(I + (1 - t)K + tK')$ for $t \in [0, 1]$. The matrix $K(t) = (1 - t)K + tK'$ is PSD for all t (convex combination of PSD matrices), so $I + K(t) \succ 0$ and g is well-defined. Differentiating:

$$g'(t) = \text{tr}((I + K(t))^{-1}(K' - K)).$$

By Cauchy–Schwarz for the trace inner product, $|\text{tr}(AB)| \leq \|A\|_F \|B\|_F$. Since the eigenvalues of $(I + K(t))^{-1}$ lie in $(0, 1]$:

$$\|(I + K(t))^{-1}\|_F \leq \sqrt{N}.$$

Therefore $|g'(t)| \leq \sqrt{N} \cdot N\alpha = N^{3/2}\alpha$, and integrating:

$$|\mathcal{C}' - \mathcal{C}| = |g(1) - g(0)| \leq N^{3/2}\alpha = 4N^{3/2}\lambda\eta(D_{\max} + \eta). \quad \square$$

Theorem 4.18 (Lipschitz continuity in data). *Suppose the velocity field $v(z, t) = -\dot{\sigma}(t)\sigma(t) s(z, t)$ of the probability flow ODE (7) is $\ell(t)$ -Lipschitz in z uniformly. Let $\mathcal{X}' = \{x'_1, \dots, x'_N\}$ with $\delta := \max_i \|x'_i - x_i\|$ and define $\Lambda := (\sum_{k=1}^L w_k e^{2 \int_0^{t_k} \ell(\tau) d\tau})^{1/2}$. Then:*

$$|\mathcal{C}(\mathcal{X}') - \mathcal{C}(\mathcal{X})| \leq 4N^{3/2}\lambda\Lambda\delta(D_{\max} + \Lambda\delta). \quad (26)$$

In particular, $\mathcal{C}$ is Lipschitz in the data with constant $4N^{3/2}\lambda\Lambda D_{\max}$ for small perturbations.

Proof. By the Gronwall inequality for ODEs with Lipschitz velocity fields, the flow map $x \mapsto z_x(t)$ satisfies $\|z_{x'}(t) - z_x(t)\| \leq e^{\int_0^t \ell(\tau) d\tau} \|x' - x\|$. The embedding perturbation is therefore:

$$\|\Phi(x'_i) - \Phi(x_i)\|^2 = \sum_{k=1}^L w_k \|z_{x'_i}(t_k) - z_{x_i}(t_k)\|^2 \leq \sum_{k=1}^L w_k e^{2 \int_0^{t_k} \ell(\tau) d\tau} \delta^2 = \Lambda^2 \delta^2.$$

Setting $\eta = \Lambda\delta$ in Lemma 4.17 gives the result. $\square$

Theorem 4.19 (Stability under score approximation). *Let $\mathcal{C}_\theta$ be the complexity computed using ODE trajectories driven by an approximate score s_θ in place of the true score s . The approximate trajectory $z_x^\theta(t)$ solves (7) with s_θ replacing s . Suppose the velocity field is $\ell(t)$ -Lipschitz and $\varepsilon := \max_{x_i, t \in [0, t_L]} \|s_\theta(z_{x_i}(t), t) - s(z_{x_i}(t), t)\|$. Then the trajectory perturbation is bounded by $\|z_{x_i}^\theta(t_k) - z_{x_i}(t_k)\| \leq \varepsilon \int_0^{t_k} \dot{\sigma} \sigma e^{\int_\tau^{t_k} \ell} d\tau =: \eta_k$, and:*

$$|\mathcal{C}_\theta(\mathcal{X}) - \mathcal{C}(\mathcal{X})| \leq 4N^{3/2}\lambda\eta(D_{\max} + \eta), \quad \eta = \left(\sum_{k=1}^L w_k \eta_k^2\right)^{1/2}. \quad (27)$$

Moreover, $\mathcal{C}_\theta$ retains all algebraic properties exactly (PSD kernel, bounded monotonicity, sub-additivity, submodularity), since every proof depends only on Φ_θ being a map into a Euclidean space.

Proof. The trajectory perturbation bound follows from the standard Gronwall comparison for ODEs with perturbed right-hand sides. Setting $\eta = (\sum_k w_k \eta_k^2)^{1/2}$ in Lemma 4.17 gives the bound. The algebraic properties hold because the proofs of Theorems 4.3–4.15 use only that Φ maps into a Euclidean space and the kernel is the Gaussian RBF applied to Euclidean distances; the same structure holds verbatim for Φ_θ . $\square$

5 Practical Considerations

5.1 Computational complexity

Computing $\mathcal{C}(\mathcal{X})$ involves three stages:

1. **Trajectory embedding** (N ODE solves). For each data point x_i , solve the probability flow ODE (7) from $t = 0$ to $t = t_L$, recording $z_{x_i}(t_k)$ at the L evaluation points. Each ODE solve requires $\mathcal{O}(S)$ score evaluations, where S is the number of integration steps (typically $S = 20$ – 100 for an adaptive Runge–Kutta method). Total cost: $\mathcal{O}(NS \cdot C_{\text{fwd}})$, where C_{fwd} is the cost of one forward pass of the score network. This is embarrassingly parallel over i .

2. **Kernel matrix assembly** (N^2 kernel evaluations). Compute $D(x_i, x_j)^2 = \|\Phi(x_i) - \Phi(x_j)\|^2$ for all pairs and exponentiate: $K_{ij} = \exp(-\lambda D^2)$. Pairwise distances in $\mathbb{R}^{DL}$ cost $\mathcal{O}(N^2 \cdot DL)$. Memory: $\mathcal{O}(N^2)$ for the kernel matrix.
3. **Log-determinant** ($\mathcal{O}(N^3)$). Compute the Cholesky factorisation $I + K = LL^\top$ (guaranteed to succeed since $I + K \succ 0$) and sum the logarithms of diagonal entries: $\mathcal{C} = 2 \sum_i \log L_{ii}$.

The bottleneck for large N is the $\mathcal{O}(N^3)$ Cholesky step and the $\mathcal{O}(N^2)$ memory for K . For datasets where N exceeds $\sim 10^4$, one may employ:

- **Nystrom approximation:** approximate K using a random subset of $m \ll N$ landmark points, reducing the factorisation to $\mathcal{O}(m^2 N)$.
- **Random Fourier features:** approximate the Gaussian kernel via a finite-dimensional random feature map, giving an explicit low-rank factorisation of K .
- **Stochastic trace estimators:** use Hutchinson-type estimators to approximate $\log \det(I + K)$ without forming K explicitly.

5.2 Choice of hyperparameters

The measure $\mathcal{C}$ depends on three hyperparameters: the kernel bandwidth λ , the time grid $\{t_1, \dots, t_L\}$, and the time weights $\{w_1, \dots, w_L\}$.

Bandwidth λ . The parameter λ sets the resolution of the similarity function. It has a geometric interpretation: $1/\sqrt{\lambda}$ is the characteristic distance in trajectory space below which two points are considered “similar” (kernel value $\geq e^{-1}$).

Crucially, λ must not depend on the dataset $\mathcal{X}$ whose complexity is being measured. If λ is chosen adaptively (e.g., via the median of the pairwise score distances within $\mathcal{X}$), the kernel matrix is implicitly re-normalised for each input, and the resulting complexity values are not comparable across datasets: a structurally richer dataset may receive a larger λ that compresses its kernel spectrum, yielding the same $\mathcal{C}$ as a simpler one.

Instead, λ should be *calibrated once* from an independent reference sample $\mathcal{X}_{\text{ref}}$ drawn from the model’s training distribution p_{data} . Concretely, one draws N_{ref} points from p_{data} , computes their trajectory embeddings, and sets λ via the median heuristic on that reference sample:

$$\lambda = \frac{1}{\text{median}\{D(x_i^{\text{ref}}, x_j^{\text{ref}})^2 : i < j\}}. \quad (28)$$

This makes λ a property of the score model and the data distribution—not of the particular dataset being evaluated. All complexity evaluations under the same model then use the same fixed λ , ensuring comparability.

Very large λ (small bandwidth) makes all off-diagonal kernel entries negligible, so $K \approx I$ and $\mathcal{C} \approx N \log 2$ regardless of the data’s structure. Very small λ (large bandwidth) makes all entries close to 1, so $K \approx \mathbf{1}\mathbf{1}^\top$ and $\mathcal{C} \approx \log(1 + N)$ regardless of diversity. Intermediate values of λ yield the most informative complexity estimates.

Time grid $\{t_1, \dots, t_L\}$ and weights $\{w_1, \dots, w_L\}$. The time points determine which stages of the ODE trajectory contribute to the distance. Very small t (trajectory barely moved from x) carries mainly positional information; very large t (trajectory near global mean) carries little discriminative information. Intermediate times are most informative: they capture the regime where within-cluster noise has been reduced by the ODE contraction but between-cluster separation persists.

In practice, the grid is typically chosen as a uniform or geometric sequence in $[0, 1]$. The number of points L controls the resolution of the multi-scale representation; L in the range 10–50 usually suffices. The weights w_k can be set uniformly or tuned to emphasise intermediate scales. A natural data-driven choice is $w_k \propto \text{Var}[\|z_{x_i}(t_k) - z_{x_j}(t_k)\|^2]$ over the reference sample, which upweights time points where trajectory distances are most informative.

Sensitivity. By Theorem 4.18, $\mathcal{C}$ is Lipschitz in the data and hence varies smoothly as a function of the hyperparameters (which enter through D). The main qualitative features of $\mathcal{C}$ (which clusters are resolved, the ranking of datasets by complexity) are typically robust to moderate changes in λ and the scale grid.

5.3 Score approximation in practice

Theorem 4.19 bounds the effect of using an approximate score s_θ in place of the true score s . The approximation error propagates through the ODE trajectory: a small pointwise score error ε at each step accumulates along the trajectory, but is controlled by the Gronwall bound.

In the denoising score matching framework (6), the training objective directly minimises $\mathbb{E}[\|s_\theta(X_t, t) - s(X_t, t)\|^2]$, so the score matching loss provides a natural proxy for ε . Well-trained diffusion models achieve low score matching loss, suggesting that $\mathcal{C}_\theta$ is a good approximation to $\mathcal{C}$ in practice.

Crucially, all algebraic properties (PSD, monotonicity, sub-additivity, submodularity) hold *exactly* for $\mathcal{C}_\theta$, regardless of the quality of the score approximation. The approximation error affects only the *value* of $\mathcal{C}$, not its structural properties.

6 Experiments

We verify the theoretical properties of $\mathcal{C}$ on synthetic Gaussian mixture models (GMMs), where the score function is available in closed form. All experiments use a variance-preserving (VP) noise schedule with $\beta(t) = \beta_{\min} + (\beta_{\max} - \beta_{\min})t$, $\beta_{\min} = 0.1$, $\beta_{\max} = 20$, and record the probability flow ODE trajectory at $L = 20$ uniformly spaced time points in $[0.01, 0.99]$.

The kernel bandwidth λ is calibrated once from an independent reference sample (a single unit-variance Gaussian) using the median heuristic, and held fixed across all experiments. This ensures that λ captures within-cluster scale and is not influenced by between-cluster separation.

6.1 Growth rate

We test the two asymptotic regimes identified in Section 3.9.

Maximally diverse regime. N points are drawn one per cluster from N well-separated clusters (placed along a line with spacing 20, each with unit covariance in $\mathbb{R}^2$). The model is an N -component GMM with equal weights. The measured complexity matches $N \log 2$ to three decimal places (Figure ??), confirming Proposition ?? (iii).

Redundant regime. N points are drawn from a single component of a fixed 3-component GMM. Two sampling variances are compared: $\sigma = 0.1$ (points nearly identical in embedding space) and $\sigma = 1$ (within-cluster diversity is resolved by the kernel). At $\sigma = 0.1$, the complexity closely follows $\log(1+N)$, as predicted by Proposition ?? (i). At $\sigma = 1$, the growth is intermediate between the logarithmic and linear regimes, reflecting the fact that each new point contributes genuine information but less than a point from an independent cluster would.

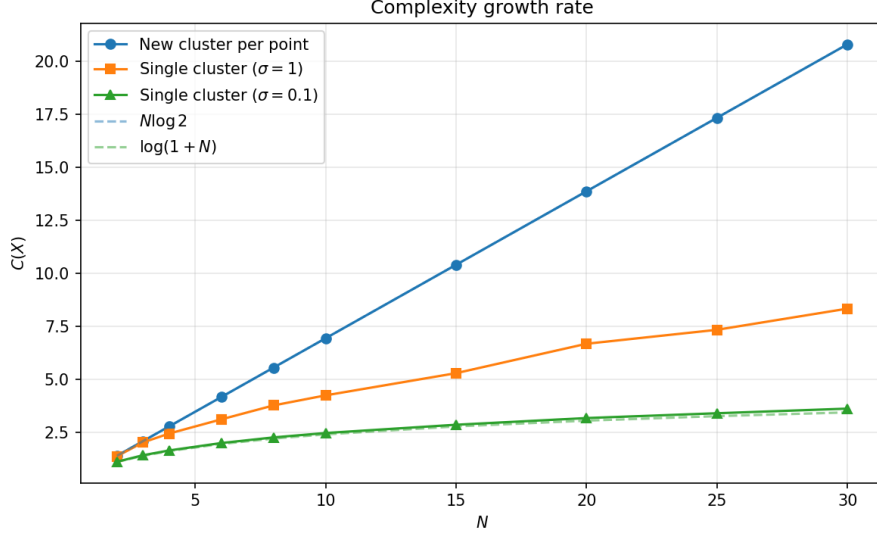


Figure 1: Complexity $\mathcal{C}$ vs. sample size N . Blue: one point per cluster (N clusters), matching the theoretical upper bound $N \log 2$. Green: single cluster with $\sigma = 0.1$, matching the theoretical lower bound $\log(1 + N)$. Orange: single cluster with $\sigma = 1$, showing intermediate growth where within-cluster diversity is resolved by the kernel.

6.2 Sensitivity to cluster separation

We fix $K = 3$ clusters with unit covariance in $\mathbb{R}^2$ and draw $N = 30$ points (10 per cluster, round-robin). The cluster means are placed on a circle of radius δ , and δ is varied from 0.5 to 12.

Figure ?? shows that $\mathcal{C}$ increases monotonically with separation and plateaus once clusters are fully resolved ($\delta \gtrsim 5$). The plateau corresponds to the regime where between-cluster kernel entries are effectively zero and the kernel matrix becomes block-diagonal, so that $\mathcal{C} \approx K \cdot \mathcal{C}_{\text{single cluster}}$.

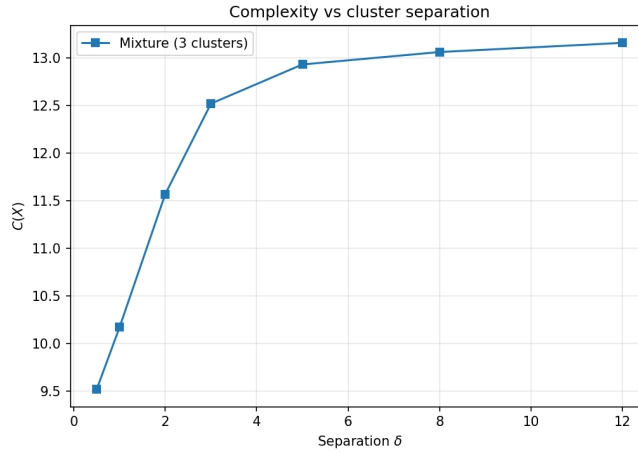


Figure 2: Complexity of a 3-cluster mixture vs. cluster separation δ , with fixed λ . The measure rises as clusters become distinguishable and plateaus once between-cluster kernel entries vanish.

7 Discussion

We have proposed a dataset complexity measure $\mathcal{C}(\mathcal{X}) = \log \det(I + K)$ that leverages the multi-scale geometry learned by score-based diffusion models through the probability flow ODE. The measure has several attractive features:

- **Grounded in learned geometry.** The trajectory distance D is derived from the probability flow ODE of a pretrained generative model, so it reflects the semantic structure of the data as learned by modern deep networks, rather than relying on hand-crafted features or ambient-space distances.
- **Cluster-aware.** Unlike the simpler score embedding (which evaluates scores at fixed data positions and loses cluster identity for well-separated clusters), the trajectory embedding preserves cluster structure because the ODE trajectories carry the full positional history of each data point through the diffusion process.
- **Provably well-behaved.** The full chain of properties—PSD kernel, bounded monotonicity, sub-additivity, submodularity, Lipschitz stability—is proved from first principles. The proofs rely only on the fact that Φ maps into a Euclidean space and the kernel is a Gaussian RBF, making them robust to the specific choice of generative model or embedding.
- **Interpretable asymptotics.** The growth rate of $\mathcal{C}$ distinguishes trivially redundant data ($\Theta(\log N)$) from maximally diverse data ($\Theta(N)$), with intermediate regimes characterised by the number of independent semantic clusters.
- **Practical applicability.** Given any pretrained diffusion model, $\mathcal{C}$ can be computed via standard ODE solving and linear algebra. All algebraic properties hold exactly even when using an approximate score function.

Limitations and future work. The $\mathcal{O}(N^3)$ computational cost limits direct application to very large datasets; scalable approximations (Nyström, random features) deserve systematic study. The choice of hyperparameters (scale grid) affects the absolute value of $\mathcal{C}$, and principled selection criteria for the noise-level grid remain to be developed. Empirical validation on standard benchmarks—comparing $\mathcal{C}$ with existing complexity proxies and correlating it with downstream model performance—is the natural next step.